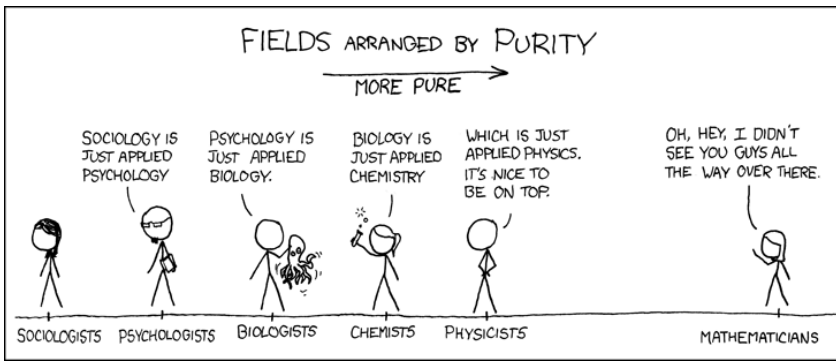
Session - 6

# AWS Deep Dive

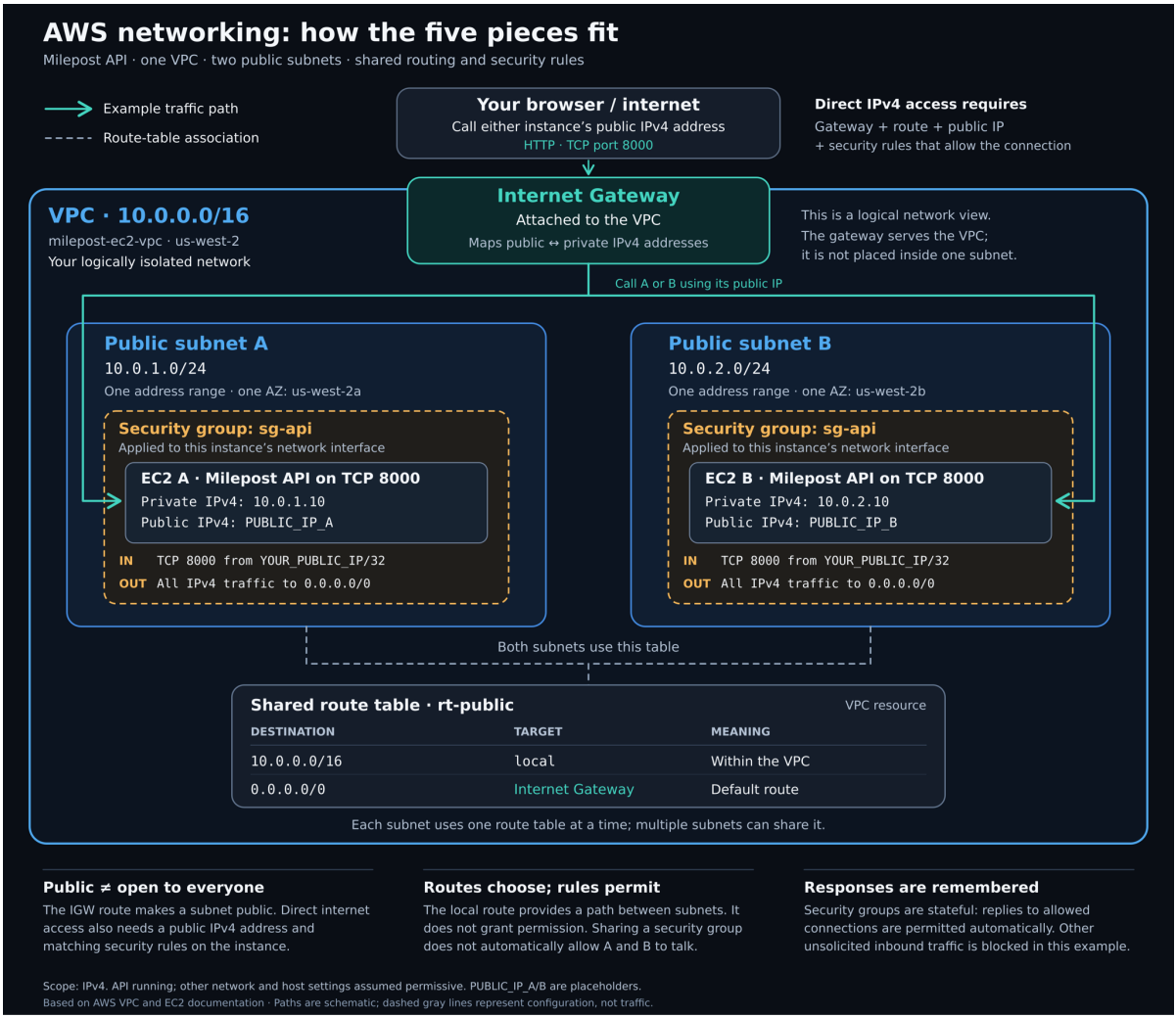
Oct 12, 2026



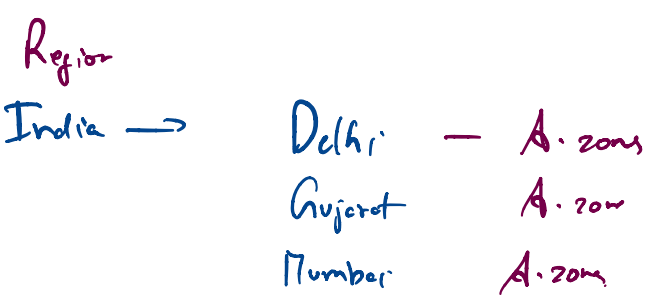




Agenda:

1. Region and Availability Zones
2. VPC
3. Subnets
4. Internet Gateway + Route Tables
5. Security Groups




VPC can span across multiple availability zones. VPC can't span across multiple regions.



A region is a physical location.

Physical data center.

Inside each region, we have multiple availability zones depending on the number of data centers available. This can vary from region to region.

A VPC can span across an entire region across multiple availability zones or a user can also create multiple VPC inside the same region.

A subnet is a smaller network which is limited to one availability zone.

Oregon has right now four availability zones. But I don't need to use all the availability zones. If I want to use only two availability zones, then I will have only two subnets.

IPV4 0-255 . 0-255 . 0-255 . 0-255

8bit 8bit 8bit 8bit

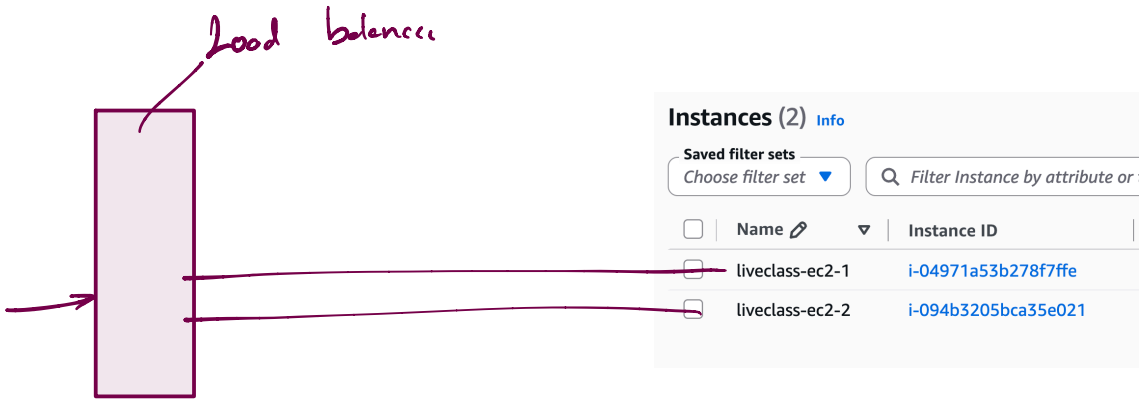
189. 32 . 41. 16




8 + 8

1 1

255 255 = 65k




1. Get/Create Access Key, and Secret Access Key from IAM (Has to be done Manually, once)
2. Containerize our image and push to ECR (Need to create a repo in ECR, pushing can be automated)
3. Create Cluster in ECS (This has to be done manually, but only once, in the beginning)
4. Create Task Definition (This will be automated)
5. Assign task to cluster and deploy (This will be automated)